

Lab 01: Text Classification – Analysis Report

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Model Comparison Table

Model	Accuracy	Precision	Recall	F1-score
Multinomial Naive Bayes (TF-IDF)	86.70%	86.66%	86.71%	86.54%
Logistic Regression (TF-IDF)	86.10%	86.11%	86.13%	86.00%
Feedforward Neural Network (PyTorch)	86.90%	86.96%	86.95%	86.89%

1. Classical vs. Neural

The feedforward network did slightly better than both classical models: 86.90% accuracy, against 86.70% for Naive Bayes and 86.10% for Logistic Regression. The gap is small because all three use the same TF-IDF input, and TF-IDF already separates these topics well using simple word counts. The network can combine features in non-linear ways, which gives it a small edge, but it cannot add information that was never in the TF-IDF vectors to begin with.

2. Naive Bayes vs. Logistic Regression

Naive Bayes beat Logistic Regression here, 86.70% versus 86.10%. This surprised me, since Logistic Regression usually does better. With only 4,000 training examples and 5,000 features, Logistic Regression may be slightly overfitting its larger set of learned weights. Naive Bayes also does well because each topic uses fairly distinct words, so its independence assumption is not a big problem on this dataset.

3. Error Analysis

Most errors from the FFNN happen between Business and Sci/Tech: 33 Business articles were labeled Sci/Tech, and 21 Sci/Tech articles were labeled Business. Sports was almost never confused with anything else. This makes sense, since articles about tech companies, like earnings or product launches, share words with both the Business and Sci/Tech categories.

4. Limitations of Bag-of-Words

TF-IDF only counts words and ignores order and context. For example, "Apple shares climb after strong earnings" mixes tech and finance words, and TF-IDF cannot tell that "Apple" is a company here, not a fruit. Word embeddings or Transformers would fix this, since they use context to tell what a word actually means in a sentence.

5. Looking Ahead

The overall pipeline, loading data, extracting features, training, and evaluating, should stay the same. What changes is the feature type: dense word embeddings instead of sparse TF-IDF counts. I expect this to help most with the Business/Sci-Tech mix-ups, since embeddings capture word meaning instead of just counting words.